

Semantic News and Oil Price Database Project

This project turns oil-market, geopolitical risk, semantic news, and country exposure CSV files into a reproducible analytics workflow with MySQL tables, SQL views, and a transparent pricing model.

4,047

daily market rows

66,660

country-month GPR rows

1.516

test RMSE, USD

One workflow, three outputs

The source CSVs become a queryable database, analysis surfaces, and predictive model artifacts.

1 Datasets

Semantic news, market prices, events, country impact.

3 Views

Daily oil/news features and event/country analysis.

2 MySQL

Operational source tables plus dim/fact reporting tables.

4 Model

StandardScaler -> Ridge for next-day Brent.

The data mixes market prices with semantic risk

Coverage spans oil prices, GPR/news indices, events, and country-level exposure.

ops_market_daily 4,047

Daily Brent/WTI, macro indicators, volatility, lags, and events.

ops_gpr_daily 15,078

Daily geopolitical risk/news index and article counts.

ops_gpr_country_monthly 66,660

Country-month geopolitical risk measures.

ops_events 55

Named geopolitical, sanctions, disaster, and market events.

ops_countries 18

Country dimension for impact and petrol analysis.

2010-2026

market table date span

1985-2026

daily GPR date span

The modeling table aligns Brent, WTI, volatility, GPR, event flags, and lag features for supervised learning.

Two database layers keep analysis flexible

Operational tables preserve source shape; dimensional tables support BI-style reporting.

Operational layer

- ops_market_daily
- ops_gpr_daily and monthly
- ops_events
- ops_countries and petrol snapshots

Dimensional layer

- dim_date, dim_country, dim_event
- fact_market_daily
- fact_gpr_daily and monthly
- fact_country_impact and petrol prices

Views turn raw tables into analysis surfaces

Three MySQL views package the joins needed for modeling, event study, and country impact analysis.

vw_daily_oil_news_features

Daily Brent/WTI, GPR, event, and volatility features for analysis and model inputs.

vw_event_price_reaction

Event-date oil price and risk indicators to inspect geopolitical market reactions.

vw_country_petrol_impact

Country exposure, petrol price snapshots, and vulnerability indicators in one reporting surface.

A transparent scikit-learn model predicts next-day Brent

Chronological split, standardized features, regularized linear model, exported metrics.

3,236

training rows

810

test rows

0.1

Ridge alpha

- Train through 2022-12-21; test through 2026-03-12
- Scale market, risk, volatility, lag, and event features
- Fit Ridge regression and save as a joblib pipeline
- Export JSON metadata and CSV test predictions

Features combine prices, risk, volatility, and events

The model uses 20 fields from ops_market_daily.csv.

2

current oil prices

2

macro indicators

1

GPR risk index

2

daily returns

6

lagged prices

4

volatility fields

1

Brent-WTI spread

2

event features

The model narrowly beats a strong baseline

Next-day oil pricing is hard because yesterday's price is already an excellent predictor.

MAE	1.118	Model error in USD; baseline is 1.113.
-----	-------	--

95.90

latest next-day Brent prediction, USD

The result is credible because it is evaluated on a future holdout period, not a shuffled split.

RMSE	1.516	Model error in USD; baseline is 1.536.
------	-------	--

MAPE	1.463%	Percent error on the chronological test set.
------	--------	--

R-squared	0.967	Variance explained on future holdout rows.
-----------	-------	--

Baseline = previous trading-day Brent price.

The handoff is reproducible

Everything needed to load, train, predict, and present is in the workspace.

Run path

- `python -m pip install -r requirements.txt`
- `python src\load_mysql.py --replace`
- `python src\train_oil_model.py`
- `python src\predict_oil_price.py`

Deliverables

- MySQL loader and views
- Database schema guide
- Trained sklearn model
- Markdown/PDF report
- Editable deck source, PPTX, PDF, and HTML